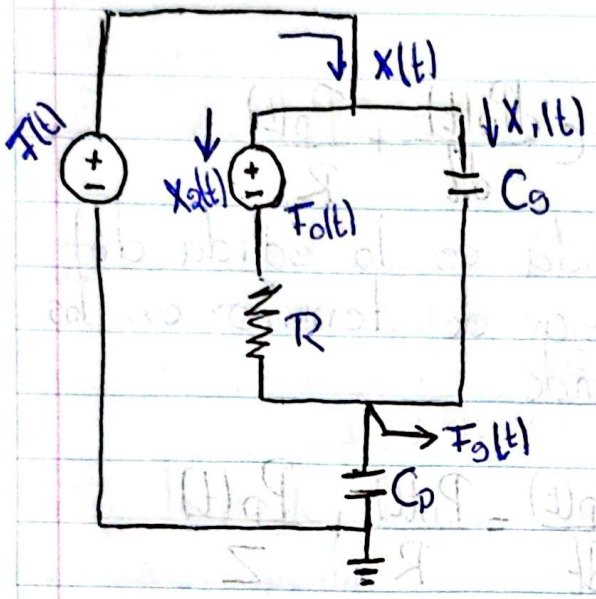
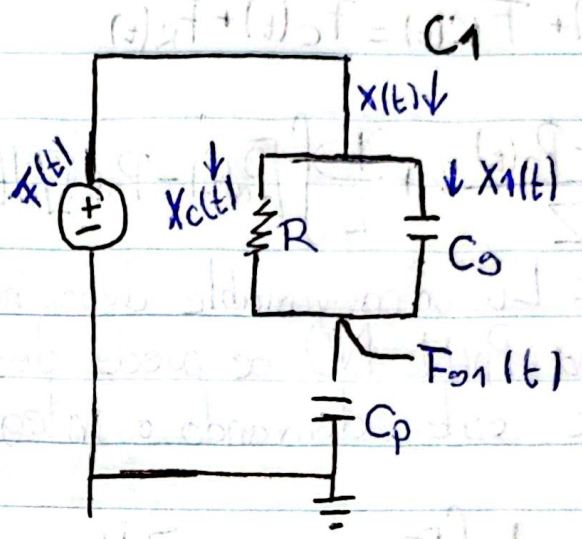
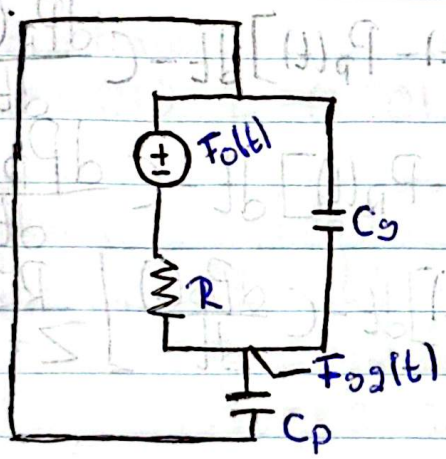
Modelado de sistemas fisiologicos

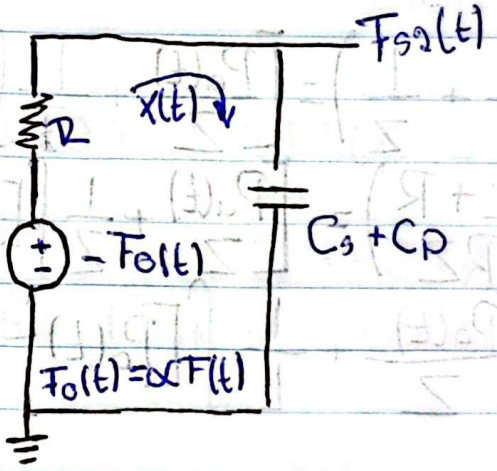
Modelo musculo esqueletico

$$F_s(t) = F_{sg}(t) + F_{sa}(t)$$









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Analisis del circuito por superposicion

Circuito 1: Apagando $F_o(t)$

$$x(t) = x_1(t) + x_2(t)$$
$$x(t) = C_p \cdot \frac{dF_{o1}(t)}{dt}$$
$$x_1(t) = C_s \frac{dF(t) - F_{o1}(t)}{dt}$$
$$x_2(t) = \frac{F(t) - F_{o1}(t)}{R}$$

$$C_p \frac{dF_{o1}(t)}{dt} = C_s \frac{dF(t) - F_{o1}(t)}{dt} + \frac{F(t) - F_{o1}(t)}{R}$$

Transformada de Laplace

$$C_p s F_{o1}(s) = C_s s[F(s) - F_{o1}(s)] + \frac{F(s) - F_{o1}(s)}{R}$$

$$C_p s F_{o1}(s) = C_s s[F_{o1}(s) + F_s(s)] + \frac{F(s)}{R} - \frac{F_{o1}(s)}{R}$$

$$\left(C_p s + C_s s + \frac{1}{R}\right) F_{o1}(s) = \left(C_s s + \frac{1}{R}\right) F(s)$$

$$\frac{R(C_p + C_s)s + 1}{R} F_{o1}(s) = \frac{C_s R s + 1}{R} F(s)$$

$$F_{o1}(s) = \frac{\frac{C_s R s + 1}{R} F(s)}{\frac{R(C_p + C_s)s + 1}{R}} = \frac{C_s R s + 1}{R(C_p + C_s)s + 1} F(s)$$

Circuito 2: Apagando $F(t)$

$$X(t) = X_1(t)$$
$$X(t) = \frac{-F_o(t)}{R}$$
$$X_1(t) = C_s + C_p \frac{dF_o(t) - F_{o2}(t)}{dt}$$

$$C_s + C_p \frac{dF_o(t) - F_{o2}(t)}{dt} = \frac{-F_o(t)}{R}$$

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Transformada de Laplace

$$C_{ps} + C_{ss} s\left[-F_o(s) - F_{s2}(s)\right] = \frac{-F_o(s)}{R}$$

$$(C_{ps} + C_{ss}) F_{s2}(s) = \left(\frac{1}{R} + C_{ps} + C_{ss}\right) - F_o(s)$$

$$-\alpha F(t) = R_x(t) + \frac{1}{C_p + C_s}\int x(t)\,dt$$

$$F_{s2}(t) = \frac{1}{C_p + C_s}\int x(t)\,dt$$

$$-\alpha F(s) = R_x(s) + \frac{X(s)}{(C_p + C_s)s}$$

$$F_{s2}(s) = \frac{X(s)}{(C_p + C_s)s}$$

$$-\alpha F(s) = -\frac{1}{\alpha}\left[R x(s)\right]$$

$$F(s) = -\frac{X(s)}{\alpha}\,\frac{R(C_p + C_s)s + 1}{(C_p + C_s)s}$$

$$\frac{F_{s2}(s)}{F(s)} = \frac{\frac{X(s)}{(C_p + C_s)s}}{-\frac{X(s)}{\alpha}\,\frac{R(C_p + C_s)s + 1}{(C_p + C_s)s}} = -\frac{\alpha}{R(C_p + C_s)s + 1}$$

$$F_{s2}(s) = -\frac{\alpha F(s)}{R(C_p + C_s)s + 1}$$

$$F_o(s) = F_{s1}(s) + F_{s2}(s) = \frac{(1 + C_s R s + 1}{R(C_p + C_s)s + 1} F(s) - \frac{\alpha F(s)}{R(C_p + C_s)s + 1}$$

$$F_s(s) = \frac{C_s R s + 1 - \alpha}{R(C_p + C_s)s + 1} F(s)$$

$$\frac{F_s(s)}{F(s)} = \frac{C_s R s + 1 - \alpha}{R(C_p + C_s)s + 1}$$

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